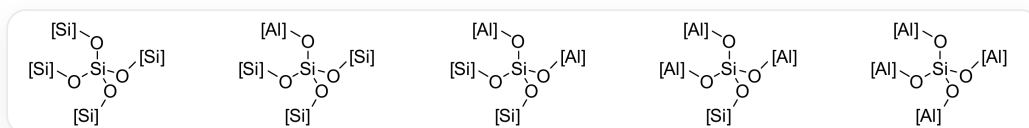


## Question

Na<sup>+</sup> ions and water molecules may exist in the cages of Y - type zeolites, and their content varies with preparation conditions. The chemical formula of a certain artificially prepared Y-type zeolite can be written as **Double subscripts: use braces to clarify**. The following determination results are known:

(1) Complete dehydration of more than 100.0 g of zeolite sample yields 72.6 g of anhydrous solid **Double subscripts: use braces to clarify**;

(2) <sup>29</sup>Si-NMR measurements show that there are five silicons with different chemical shifts in the crystal, and the ratio of their contents is:



Silicon-oxygen tetrahedron connected to four silicon atoms, silicon-oxygen tetrahedron connected to three silicon atoms and one aluminum atom, silicon-oxygen tetrahedron connected to two silicon atoms and two aluminum atoms, silicon-oxygen tetrahedron connected to one silicon atom and three aluminum atoms, and silicon-oxygen tetrahedron connected to four aluminum atoms, from left to right.

1.24 : 10.46 : 14.24 : 6.30 : 0.60

Which of the following options are incorrect:

1.  $\frac{m+x}{n} > 2$

2.  $m \times n < 0.3$

3.  $\frac{x-m}{n} < 1.4$

4.  $\frac{m}{n} > 0.4$

5.  $\frac{n-m}{x} > 0.32$

6.  $\frac{m}{x} < 0.25$

7.  $n > \frac{x}{1.8}$

8.  $m \times x > 0.40$

A. 2,6

B. 1,2,6,8

C. 3,5,7

D. 2,3,5,7

E. 1,2,6

F. 2,3,5,7,8

G. 1,2,3,4,5,6,7,8

H. 3,4,7

I. 1,2,4,6

J. All of the above options are incorrect or the answer is incomplete.

## Answer

Correct Answer: C

## Detailed Explanation

Lowenstein's rule states that in aluminosilicates (such as zeolites), two aluminum-oxygen tetrahedra cannot be connected by a corner, that is, there is no  $\text{Al} - \text{O} - \text{Al}$  oxygen bridge.

### CHECKPOINT

1 PTS

there is no  $\text{Al} - \text{O} - \text{Al}$  oxygen bridge

The relative content of Si in the crystal is:  $1.24 + 10.46 + 14.24 + 6.30 + 0.60 = 32.84$

### CHECKPOINT

1 PTS

The relative content of Si is: 32.84

The relative content of Al in the crystal is:  $\frac{10.46+2 \times 14.24+3 \times 6.30+4 \times 0.60}{4} = 15.06$

### CHECKPOINT

1 PTS

The relative content of Al is: 15.06

According to charge conservation,  $m + n = 1$

### CHECKPOINT

1 PTS

$$m + n = 1$$

Therefore:  $n = \frac{32.84}{32.84+15.06} = 0.686$

### CHECKPOINT

1 PTS

$$n = 0.686$$

$$m = 1 - n = 0.314$$

### CHECKPOINT

1 PTS

$$m = 0.314$$

From:  $\frac{M(\text{Na}_{0.314}[\text{Al}_{0.314}\text{Si}_{0.686}\text{O}_2])}{M(\text{Na}_{0.314}[\text{Al}_{0.314}\text{Si}_{0.686}\text{O}_2] \cdot x\text{H}_2\text{O})} = \frac{72.6}{100.0}$

We get:  $x = 1.40$

### CHECKPOINT

1 PTS

$$x = 1.40$$

Substituting into the calculation:

Option 1,  $\frac{m+x}{n} = 2.50 > 2$ , correct

Option 2,  $mn = 0.215 < 0.3$ , correct

Option 3,  $\frac{x-m}{n} = 1.58 > 1.4$ , incorrect

Option 4,  $\frac{m}{n} = 0.458 > 0.4$ , correct

Option 5,  $\frac{n-m}{x} = 0.266 < 0.32$ , incorrect

Option 6,  $\frac{m}{x} = 0.224 < 0.25$ , correct

Option 7,  $\frac{x}{1.8} = 0.778 > n$ , incorrect

Option 8,  $mx = 0.440 > 0.4$ , correct

In summary, options 3, 5, and 7 are incorrect, so the correct answer is C.

**CHECKPOINT**

1 PTS

Options 3, 5, and 7 are incorrect, so the correct answer is C